Practical Assignment for Fast Learner

College Course & Student Management System

Objective

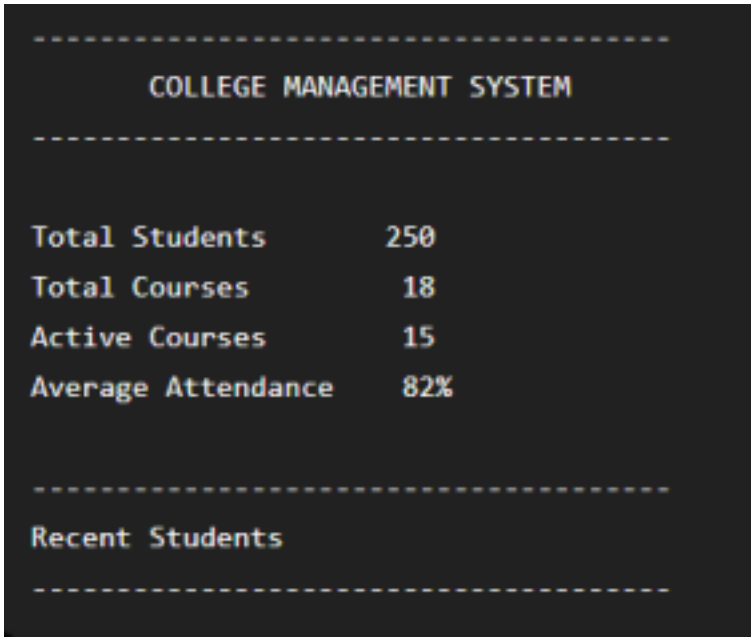
Develop a **complete College Course Management System** using **React**, incorporating reusable components, routing, forms, events, arrays, conditional rendering, CSS/Sass and performance optimization.

This assignment should combine most of the concepts from both units.

Application Modules

1. Dashboard

Display:



Use React components and arrays to generate the statistics.

2. Student Management

Create a student form:

- Student Name
- Enrollment Number

- Email
- Course
- Semester
- Gender
- Attendance

Functions:

- Add student
- Edit student
- Delete student
- Search student
- Filter by course
- Display student list

Use:

- React forms
- Events
- Props
- Arrays
- Conditional rendering

3. Course Management

Create a course list:

```
Course Code | Course Name | Faculty | Credits
-----------------------------------------------
BCS101      | Web Tech    | Dr. A   | 4
BCS102      | DBMS        | Dr. B   | 4
BCS103      | AI          | Dr. C   | 3
```

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Students should be able to:

- Add courses
- Delete courses
- Search courses
- Filter courses
- View course details

4. Student Details

Using **React Router**, create:

- /students
- /students/:id
- /courses
- /courses/:id
- /dashboard

Clicking a student should open:

```
Student Details

Name: Rahul Sharma
Enrollment: 1023
Course: B.Tech CSE
Semester: 5

Attendance: 86%

Subjects
-----------------------
Web Technology      85%
DBMS                90%
AI                  78%
```

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5. Attendance Warning

Implement conditional rendering.

For example:

Attendance >= 75%

→ Eligible

Attendance < 75%

→ Warning

Attendance < 60%

→ Detained

Example:

```
Rahul Sharma
Attendance: 82%
Status: Eligible
```

and:

```
Amit Patel
Attendance: 58%
Status: Detained
```

6. Performance Optimization

Use **React Memoization** for an appropriate component.

For example:

- StudentList
- StudentCard
- CourseList

Use React.memo() and/or useMemo() where it provides a meaningful benefit.

Students should also explain **why** memoization was used.

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7. Styling

Students must provide:

- Responsive layout
- Navigation bar
- Cards
- Tables
- Forms
- Buttons
- Alerts

They may use either:

CSS

or

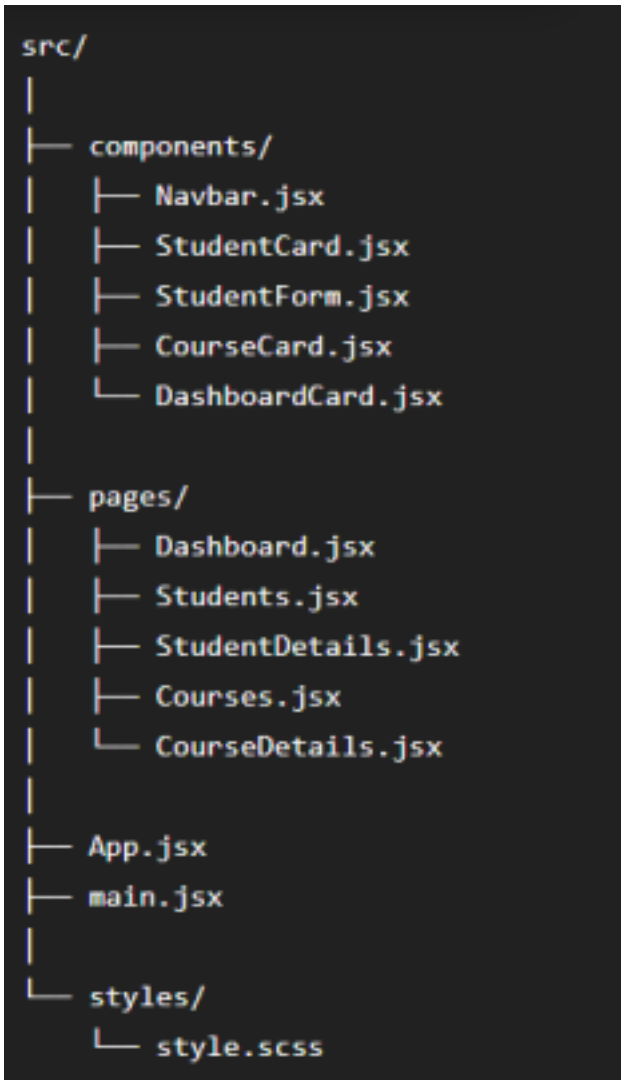
Sass/SCSS

Required React Component Structure

Students should organize the project approximately like:

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src/
components/
Navbar.jsx
StudentCard.jsx
StudentForm.jsx
CourseCard.jsx
DashboardCard.jsx
pages/
Dashboard.jsx
Students.jsx
StudentDetails.jsx
Courses.jsx
CourseDetails.jsx
App.jsx
main.jsx
styles/
style.scss